

Logistic Regression

Objective The goal of this assignment is to implement stochastic gradient decent approach for logistic regression and apply it to the task of determining whether documents are talking about hockey or baseball. You should not use any libraries that implement any of the functionality of logistic regression for this assignment; logistic regression is implemented in `scikit learn` and many other toolkits, but you should do everything by hand now. You'll be able to use library implementations of logistic regression in the future.

Coding (65 points) Understand how `logreg.py` creates feature vectors (this will help you code the solution and to do the later analysis). You don't actually need to write any code for this, however. (Optional) Store necessary data in the constructor so you can do classification later. You'll likely need to write some code to get the best/worst features (see below). Modify the `sg_update()` function to perform non-regularized updates.

Analysis (35 points) What is the role of the learning rate? How many passes over the data do you need to complete for the model to stabilize? These metrics give you clues about what the model is doing, but no one metric is perfect. Pretend you need to explain to your boss what features you use to discover a post is about baseball or hockey. How would you answer this question? What words are the best predictors of each class? How (mathematically) did you find them? What words aren't good predictors of either class? How (mathematically) did you find them?

(Optional) Extra Credit First, use a schedule to update the learning rate. To do so, supply an appropriate argument to `step` parameter, support it in your `sg_update()` function, and show the effect in your analysis report. Second, use document frequency (provided in the vocabulary file) to modify the feature values to `tf-idf`. There are many variants of `tf-idf`; you are welcome to implement any of them. Please specify in your analysis which variant you choose. Once chosen, modify `class Example` to store the `tfidf` vector in a variable named `x_tfidf`. With the appropriate flag, use the `x_tfidf` vector rather than `x` in the update, and show the effect in your analysis document. Third, modify `sg_update()` to perform lazy regularized updates, which only update the weights of features when they appear in an example. Show the effect in your analysis report.

Caution: If/When implementing extra credit, make sure your implementation of the regular algorithms doesn't change.

Example This is an example of what your runs should look like:

```
$ python logreg.py
Read in 1064 train and 133 test
Update 1 TP -795.043334 HP -93.911814 TA 0.497180 HA 0.533835
Update 6 TP -724.253354 HP -93.249903 TA 0.572368 HA 0.518797
Update 11 TP -780.935341 HP -98.223121 TA 0.549812 HA 0.466165
Update 16 TP -732.288174 HP -89.181998 TA 0.565789 HA 0.601504
...
Update 41 TP -725.644667 HP -84.026452 TA 0.661654 HA 0.631579
Update 46 TP -664.738996 HP -79.949308 TA 0.686090 HA 0.646617
Update 51 TP -619.667363 HP -76.821255 TA 0.706767 HA 0.684211
Update 56 TP -619.743899 HP -75.954954 TA 0.709586 HA 0.646617
...
```

Hints Make sure that you debug on small datasets first; I've provided toy data to get you started. Certainly make sure that you do the unregularized version first and get it to work well. Use numpy functions whenever you can to make the computation faster. For the tf-idf extra credit, use the vocab file: the second element is the document frequency. This requires you to modify the `read_dataset()` function.

Submission Guideline

Submit a zip file named `NAME_STUDENTID.zip` containing:

1. `logreg.py`: include your name at the top of the source.
2. `report.pdf`: no more than one page. To minimize effort for the grader, you'll need to put everything on a page. Take this into account when selecting if/which extra credit to do. Think of the page requirement like a regularizer. Note that pictures are better than text in reporting results.

Submit your zip file through the link on Blackboard. Submissions elsewhere won't be considered.